

# Fracture Detection In X-rays Using Custom CNN And Transfer Learning Models

Amna Hassan<sup>1</sup>; Ilsa Afzaal<sup>2</sup>; Nouman Munib<sup>3</sup>;  
Aneeqa Batool<sup>4</sup>; Hamail Noor<sup>5</sup>

<sup>1</sup>University of Engineering and Technology, Taxila



## Introduction

Bone fractures are a significant cause of global health care issues, causing pain, disability and lost productivity. Expert interpretation, high expense or risk to radiation exposure limits conventional diagnostic techniques like X-rays, CT scans and MRIs. Subtle fractures often remain undetected in X-rays, specifically in low-resource environments where radiology expertise are not readily available.

Advances in artificial intelligence, particularly in Convolutional Neural Networks, have evidenced considerable promise for fracture detection directly from imaging. Comparing to conventional methods that employ hand-engineered features, CNNs automatically learn sophisticated patterns and perform at a diagnostic level comparable to radiologists. However, most of the existing models rely on transfer learning from non-medical datasets which undermines their clinical dependency.

This research puts forward a CNN model that has been specifically tailored for fracture detection in X-rays, intending to deliver a lightweight and efficient solution for tackling these challenges.



## Methods and Materials

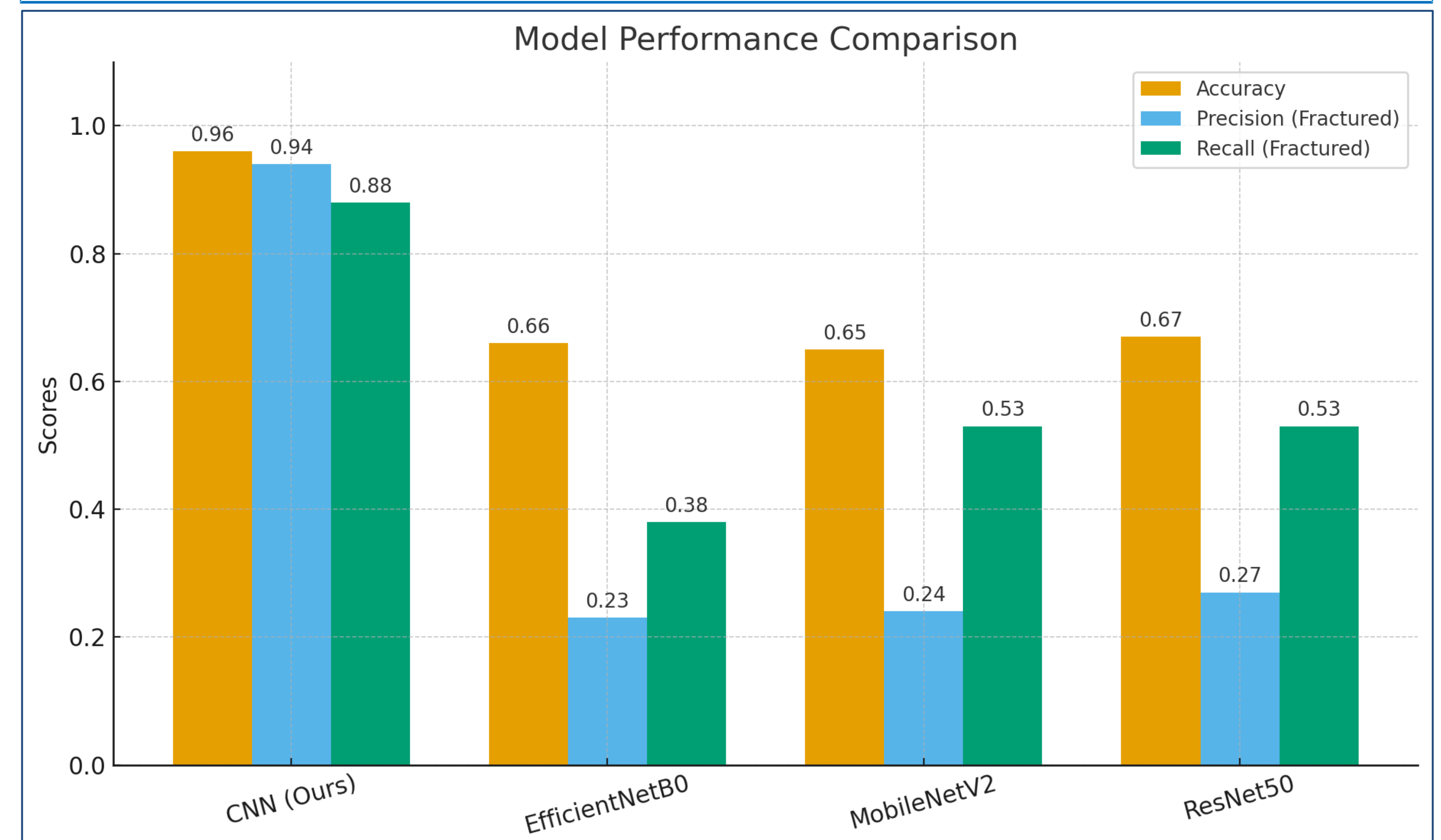
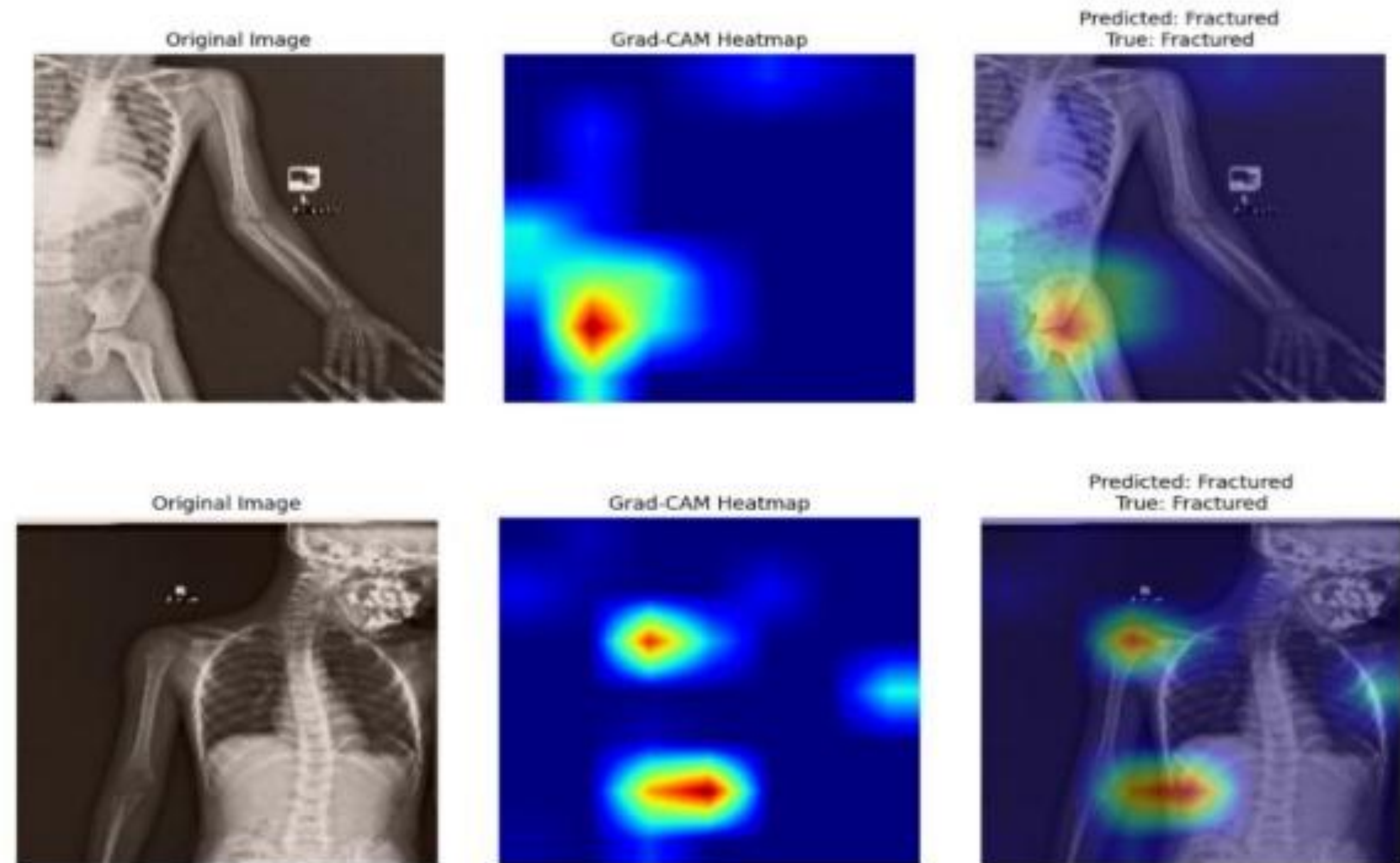
- **Dataset:** FracAtlas Dataset comprising of 4,083 X-ray images (3,366 intact, 717 fractured) encompassing leg, hip, and regions.
- **Preprocessing:** Images resized to 128×128 pixels (for custom CNN) and 224×224 pixels (for transfer learning). Data augmentation employed with rotation, zooming, and flipping and applied stratified 80/20 split for training and testing.
- **Model:** Custom-built CNN comprising of three convolutional blocks each with ReLU activation, batch normalization, max pooling, dropout, dense layers and sigmoid output aiming for binary classification, trained using Adam optimizer with binary cross-entropy loss
- **Comparisons:** Compared to EfficientNetB0, MobileNetV2, ResNet50.
- **Evaluation Metrics:** Accuracy, Precision, Recall, F1-score and confusion matrix used to evaluate models' performance. Grad-CAM visualization provides interpretability by highlighting the regions the model focuses on.
- **Training:** Custom-built CNN model trained for 30 epochs whereas transfer learning models were trained for 5 epochs, by means of Adam optimizer and binary cross-entropy loss.

## References

- [Lindsey et al., 2018] Deep neural network improves fracture detection by clinicians.
- [Ma and Lou., 2021]. Two stage CNN for fracture detection.
- [Abadeen et al., 2023]. FracAtlas dataset for musculoskeletal fracture diagnosis.
- [Cohen et al., 2023]. AI vs. radiologists in wrist fracture detection.

## Results

- The proposed CNN model achieved 95.96% accuracy, substantially higher than competing approaches.
- Transfer learning models (EfficientNetB0, MobileNetV2, ResNet50) demonstrated accuracies in the 65–67% range, underscoring their limited generalization compared to our architecture.
- Precision (Fractured): 0.94 (CNN) vs. 0.23–0.27 (transfer learning).
- Recall (Fractured): 0.88 (CNN) vs. 0.38–0.53 (transfer learning).
- Confusion matrix highlights minimal false positives (35) and manageable false negatives (71).
- Grad-CAM visualizations confirm accurate localization of fracture regions.



## Conclusion

Our research develops a custom CNN model for accurate fracture detection within X-ray images. Trained and tested on FracAtlas dataset, the model demonstrated 95.96% accuracy outclassing the transfer learning models. These results indicate that lightweight CNN model can perform robustly even for skewed medical imaging data.

### Clinical Implications:

- Potential use in resource-constrained healthcare environments.
- Decreased reliance on radiological specialists.
- Rapid detection and diagnosis of fractures.

**Future Work:** This study attests to the validity of our method highlighting the value in extending into multi-center datasets, varied imaging modalities, and clinical validation to support deployment in health-care settings.